

The GAIA Construct Grid: Role, Protocol, Substrate

Instrument ID: ERES-GAIA-GRID-2026-001

Version: v0.3 (DRAFT — pre-MIEVM; Appendix B added at v0.2; TheWall substrate closed at v0.3)

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Abstract

This instrument canonizes the mapping of the five ERES constructs — TheMall, Sentry, TheWall, Avatar, TheALL — and the enclosing Construct GAIA onto the two-sided synchronization band established in Sound Holds the Band (Theorem 1, Figure 1). The mapping follows the functions-at-boundaries principle: the five regimes of Figure 1 (collapse, upper wall, stable resonance band, lower wall, incoherence) remain physics; the constructs are roles — actors with functions, protocols, and substrates — positioned by what work they do relative to the band, not by which regime they occupy. The completed grid yields a candidate structural regularity (the Handoff Pattern), three falsifiable predictions expressible in the existing Spectral Test Battery, a canonical definition of infinence, and a provenance claim: the three pre-ERES ventures (CyberRAVE 1997, SaleBuilders 2007, GunnySack) recapitulate as functional organs of the construct. Doctrine of Verifiable Claims tags appear inline throughout.

1. Position and Provenance

This instrument extends ERES-PELLIS-SHB-001 v2.2 from physics to architecture. SHB establishes [DERIVED, per SHB Theorem 1] that coherent transmission exists only inside a two-sided band: $K_1 < K < K_2$, bounded below by the critical coupling K_c (incoherence beneath) and above by $K_{\text{condensation}}$ (over-lock, $r \rightarrow 1$, collapse beneath the appearance of success). SHB Corollary 1 establishes that full synchronization is not resonance-preserving — the formal tune-not-mine result.

The question this instrument answers: where do the ERES constructs live relative to that band? The naive answer — assign one construct per regime — fails on inspection (it condemns TheMall to a failure state, inverts TheALL, and puts three walls in a figure that defines two). The corrected answer is the functions-at-boundaries principle.

2. The Functions-at-Boundaries Principle [CONSTRUCTED]

Regimes are states of the coupling; constructs are roles acting on the coupling. A regime cannot guard itself, price itself, or hold itself open — those are functions, and functions require actors. Accordingly:

- Boundaries get keepers (active detection and enforcement at the critical couplings).
- The band gets a holder (the structure that keeps K inside (K_1 , K_2)).
- Pressures get governors (bounded schedules replacing unbounded drift).
- The substrate gets a grader (structure without pricing).
- The whole gets an enclosure (damping across all regimes — a governor cannot be inside the thing it damps).

This converts a taxonomy claim (which regime is which component) into a mechanism claim (which component does what work). Mechanism claims are reachable by the validation suite; taxonomy claims are not.

3. The Grid

Construct	Role	Protocol	Substrate	Sentry Instance
GAIA	The Construct: governor-not-judge; continuous damping across all regimes	GAIA-vote (Transmission at governance scale, per SHB §7)	— (enclosure, not layer)	— (Sentry reports to GAIA)
TheMall	Upward ascent pressure; ungoverned limit = condensation/mining ($r \rightarrow 1$)	EarnedPath — bounded ascent schedule (CPM $\times$ WBS + PERT)	Gracechain · Meritcoin [CANDIDATE]	GunnySack/RT_Media (98/2) over SaleBuilders

Sentry	Role class: keeper of the upper wall ($K_{condensation}$), instanced per subsystem	Spectral Test Battery (SHB App. A, nine tests); alarm policy 98/2	CyberRAVE (structure) · RT_Media (trajectory)	— (is itself the instance class)
TheWall	Holds the resonance band open: $K_1 < K < K_2$	SECUIR [CONSTRUCT ED]	GERP — shared-budget state layer [CONFIRMED at v0.3]	CyberRAVE-ins tanced Sentry
Avatar	Keeper of the lower wall (K_c): minimum coupling interface	PlayNAC — entry entrainment (operates from K_c upward)	Ship's-Mate (graphene wearable)	—
TheALL	Infinance substrate: undifferentiated ground state below K_c	Context Field (CF) — seven-tier awareness ladder	Bio + graphene carbon carriers [PROPOSED]	—

4. Row Commentary with DVC Tags

GAIA. [CONSTRUCTED, doctrine-consistent] GAIA is drawn as a dashed enclosure, not a band, because it is the one element that is not a layer: it is the damping operator across all five regimes. This is the governor-not-judge doctrine rendered literally.

TheMall. The construct is not the collapse regime; it is the pressure whose ungoverned limit is collapse. [CONJECTURE — TAGGED] EarnedPath is the governing protocol: $CPM \times WBS + PERT$ is a bounded ascent schedule.

Sentry. Sentry is a role class, instanced per subsystem. Its protocol is the nine-test Spectral Test Battery. Its alarm policy is 98/2. [CONSTRUCTED — TAGGED] 98/2 is a declared, versioned, per-instance-tunable default policy parameter.

TheWall. SECUIR fills the protocol slot. [CONSTRUCTED — TAGGED] Substrate: GERP (Global Earth Resource Planner) — CONFIRMED by the author at v0.3 following the three-hypothesis discrimination suite.

Avatar. Avatar is the role (keeper of K_c); PlayNAC is its protocol; Ship's-Mate is its substrate.

TheALL. The incoherence regime is reframed [CONSTRUCTED — TAGGED] as substrate: undifferentiated potential prior to coupling. Its economic face is infinance.

5. The Handoff Pattern [CANDIDATE DERIVED]

Proposition 1. Every protocol family in the grid straddles exactly one boundary and hands off to the adjacent family at a keeper.

6. Falsifiable Predictions

1. Prediction 1 (PlayNAC entry signature): Session telemetry shows order-parameter trajectories that begin below the band and enter it (rising-r signature).
2. Prediction 2 (EarnedPath band respect): EarnedPath telemetry shows stepped, plateauing r-trajectories that saturate in-band.
3. Prediction 3 (Sentry margin): A 98/2-configured Sentry instance alarms before wall contact with measurable alarm-to-breach margin $\geq 2\%$ of band width.

7. Open Items

- CF tier $\leftrightarrow$ K_c declaration: Pending.
- TheWall substrate: CLOSED at v0.3 (GERP).
- TheALL substrate: Proposed (Bio + graphene).
- SECUIR mechanism sentence: Pending.
- 98/2 derivation: PARTIALLY CLOSED ($m^* = v\tau$).

8. Definition: Infinance [CONSTRUCTED]

infinance (n.): the unpriced potential of the incoherent substrate; value below K_c — real but untransmitted; the domain from which finance draws and to which

Proof-of-Resonance is the only admissible bridge.

9. Trio Provenance

CyberRAVE (1997), GunnySack/RT_Media, SaleBuilders (2007–2012) recapitulate as functional organs of the construct.

10. Rating Log

Single-node pre-MIEVM ratings recorded.

Appendix B — Mathematical Formalization

B.1 Inherited Layer [DERIVED]

Substrate dynamics are Kuramoto. Lower wall: K_c . Upper wall: K_{cond} . Band width: ΔK .

B.2 Constructs as Operators on $K(t)$ [CONSTRUCTED]

- **TheMall:** $\dot{K} = \mu > 0$.
- **EarnedPath:** $\dot{K} = \mu(1 - K/K_\infty)$.
- **Sentry:** $m \geq v\tau$.
- **Avatar/PlayNAC:** Entry crossing $K(t^*) = K_c$.
- **Context Field:** Grading without pricing.
- **Handoff Pattern:** Closed intervals covering the coupling axis.
- **GAIA:** Continuous feedback $\dot{K} = f_{drive} - \gamma \max(0, r - r_{set})$.

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